

Report ideas

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1 Primary observations

1.1 One. - L^2 boundedness

p18 / line44 does not seem to exist but it probably refers to

$$\|u_i^\epsilon - u_i^\epsilon(\cdot + \xi, t)\| \rightarrow 0 \text{ as } |\xi| \rightarrow 0$$

from $u_i^\epsilon \in L_t^2 L_x^2$.

This holds because $C_0^\infty(\Omega)$ is dense in $L^2(\Omega)$, and for functions φ_n approaching the L^2 function $f = u_i^\epsilon$ the property holds, thus it holds for f as well.

1.2 Two. - Existence and coercivity

The author of the report in this comment states that the existence result that is given by the Temam article is for the anisotropic case, ie. not for the primitive equations. The author also says that the existence of the limit model seems to be the result of the Azerad-Guillen article.

- i) As for the first statement, the result of the Temam article actually is for the primitive equations (not for the non-hydrostatic model). Because the existence in the Temam article is for the limit model, and our goal is also for the limit model case, the coercivity part is essentially the same, given the fact that the terms are similar.
- ii) The Azerad-Guillen article yields a convergence result. A separate existence results guarantees the fact that the solutions of the anisotropic model do not converge to the empty set.

2 Secondary observations

2.1 One. - Novelty

Apart from having the result on the nonlinear part, we could add the case of a diffusion matrix.

2.2 Two. - The explanation of the boundary conditions

For the pollution term on the ground level we do not have a free boundary. It is explained in the article that while for $\partial_2 C$ and $\partial_3 C$ it is reasonable to expect to be zero, in the dominant wind direction for the sake of physical meaningfulness we decide to use the more relaxed $L^2 L^2$ boundedness criteria.

2.3 Three. - The source term

- i) Physical meaningfulness and deactivation: ok, we can look for a more meaningful source idea and check what happens. Otherwise we have a

mathematical result that is physically meaningful before and after activation, until the process lasts.

- ii) The reason why the pollution source term is introduced in two different steps is that on p6, after describing the equations describing the pollution phenomena in general, we define what the term S represents in the equation. These parts belong together. Secondly, once the equations are given, and following this, we describe the whole concept of scaling, we need to specify what happens with these source terms exactly after scaling. Essentially, the source terms are discussed before and after the scaling process - first in general, describing our choice of the delta approach, and then, secondly, more specifically, defining the term after scaling.

2.4 Four. - The impression of two-way coupling

We can make a remark either on this page, or possibly even at an earlier point, highlighting the fact the equation describing the fluid velocity field does not contain the concentration term – it is the pollution equation that includes the velocity function.

2.5 Five. - Ignoring diffusion in the downwind direction.

Assumption (6) on p7:

$$|u_1 \partial_1 C| \gg |K_x \partial_{11}^2 C| \quad (1)$$

Regarding this comment at this point we can say that our results are valid in case of a big Peclet number (ratio of advection and diffusion), which, in engineering applications is often very large. (see Peclet number). The same assumption is used in "P. Goyal and A. Kumar. Mathematical modeling of air pollutants: An application to indian urban city. In Air Quality-Models and Applications.", moreover, the following course literature which deals with rivers, oceans and atmosphere, also neglects diffusion in the streamwise direction: Dartmouth

2.6 Six. - Clarity and linguistic improvements

The approach of depth-integrating and the shallow water equations are present in the introduction since their root is the same physical / mathematical observation as the one used in the present article – even if the final concept is significantly different. They are mentioned for the sake of a fuller picture and also to highlight the fact that there are several possible paths and which direction we are not taking.